



Activity: Use generative AI to explore data visualizations



Activity overview

As a data analyst, you'll likely be tasked not only with gathering and analyzing data, but also with presenting that data to your company's teams and stakeholders in a clear and compelling fashion. The good news is that generative AI (gen AI) can help you compile reports and create data visualizations faster and more efficiently.

In this activity, you'll use a gen AI tool and the TCREI prompting framework to help you draft a report for stakeholders. This activity is optional, but we recommend it to help you level up your prompting skills. If you're unable to complete this activity, you'll still be able to progress through the course and earn a certificate from Google.

Gen AI tool access

To complete this activity, you can use the browser-based gen AI tool of your choice. The following are instructions for accessing Gemini, which requires a Google Account.

To access Gemini:

- Go to gemini.google.com.
- Sign in to your Google Account.

Refer to the resource about how to [create a Google Account](#) if you don't already have one. For more help with signing into Gemini, please refer to [Gemini Apps Help](#).

- For more details on using Gemini (such as who can use Gemini, Gemini's Privacy Notice, and where Gemini is currently available), refer to the [Gemini Apps FAQ](#).
- Don't enter private or confidential information in your Gemini conversations, or any data you wouldn't want Google to use to improve its products, services, and machine learning technologies.

Imagine you're working as a data analyst at an investment bank specializing in portfolio management. Your current task is to provide recommendations for clients on adjusting their portfolios. You've analyzed the data and are now preparing a report for key decision-makers outlining your recommendations.

One key data source you're using is the S&P 500 stock market index, a significant indicator of U.S. economic health. Your report needs to explain the significance of this data to stakeholders, but you aren't sure which type of visualization would be most helpful. To get started, you ask a gen AI tool to provide some recommendations.

As a reminder, here is the prompting framework:

- **Thoughtfully:** Describe your **task**, specifying a persona and format preference.
- **Create:** Include any **context** the gen AI tool might need to give you what you want.
- **Really:** Add **references** the gen AI tool can use to inform its output.
- **Excellent:** Next, **evaluate** the output to identify opportunities for improvement.
- **Inputs:** Then, **iterate** on your initial prompt to attain those improvements.

In this activity, you'll start prompting with just one part of the framework, then gradually build on it in order to identify what's missing or unhelpful from your prompt and make the output more useful.

Let's start by entering a simple prompt into the generative AI tool, focusing solely on the **task** at hand. We'll refine and expand this prompt using the rest of the prompting framework in subsequent steps. For now, consider this example prompt that outlines what we want the tool to do:

PROMPT: *I have a dataset with the daily closing prices for the stocks in a diversified portfolio over the course of the last year. Provide a list of the most*

applicable types of data visualizations to help me illustrate which sectors of the economy are performing best.

Remember, this initial prompt is a starting point for learning and experimentation. We'll iterate on it later to ensure it aligns with the prompting framework and produces the best possible results.

Review and assess the output from your initial prompt. Is the list of data visualizations accurate and unbiased? Does it include a variety of options? Are the data visualizations relevant to the financial industry and your report?

Identify ways that the output could better meet your needs, and use this assessment to guide your refinement and iteration process.

Revise and refine your prompt by adding more of the prompting framework:

- Add more **context** to help the tool provide better, more relevant results.
 - For example, you could:
 - Clarify what you mean by "performing best." Are you looking for highest returns, lowest volatility, or something else? This helps tailor the visualization suggestions.

- Indicate who will be viewing these visualizations. Are they financial experts or a general audience? This impacts the complexity and explanation needed.
 - Give feedback on the tone you'd like to maintain.
- Provide examples as **references**.
 - For example, you could:
 - Mention any visualizations that you liked or found insightful. This gives the AI tool a reference point for style and approach.
 - Bring up any particular sectors you want to highlight or compare. This helps focus the visualization suggestions.
 - Specify if you're interested in the entire year's performance or a specific period within the year. This impacts the data range for visualizations.
- Specify the **format** you want to receive.
 - For example, you could:
 - Request a specific number of visualization types (e.g., 3-5) to ensure a concise yet varied list.
 - Request a brief mention of the pros and cons of each visualization type to help you make informed choices.

When you're ready, enter your new prompt. Alternatively, you can build on what you've already entered by issuing follow-up requests and engaging in a dialogue with the AI tool.

Review the output again and assess what information you might still need. Are there any other parts of the prompting framework that you can include next time, or opportunities to add more detail or nuance? If so, continue to prompt.

Continue repeating steps 3 and 4 until you feel confident that the output meets your goals. For example, is the list of data visualizations long enough to give you useful choices for creating your report? Are the recommendations relevant for a financial report? Will they help you present your data in the most impactful way?

This exercise, along with the lesson videos, offer examples of how generative AI can streamline your daily tasks. Now, experiment with your preferred generative AI tool to discover how it can support your specific responsibilities as a data analyst.

^ Step 6: Questions and submissions

Answer the following questions to complete the assignment.

1. Which parts of the prompting framework did you find most useful? *Select all that apply.*

1 / 1 point

- ☐ Task (including persona and format)
- ☒ Context
- ☒ References
- ☒ Evaluate
- ☒ Iterate

2. **Submit the prompt you felt best satisfied your goal and provided enough relevant specificity, context, and references to produce a useful and effective output.**

1 / 1 point

When I give the reference, then the prompt result becomes more accurate, and whatever I need, I can get the proper output. So references produce useful and effective output.

3. **How did evaluating and iterating affect the final output? *Write 1-2 sentences.***

1 / 1 point

Evaluating and iterating are the last 2 processes; those are very vital processes. In evaluating the section, we have to check the output and scrutinize what is missing and what is needed. And in the iterating section we have to check that finally output meets the goals.